

● FEATURE & ARCHITECTURE SPECIFICATION

Library & Learning Resource Management System

A single, dynamic catalog for books, toys and educational material. Fully admin-governed, with delegated staff access, and complete visibility for parents and outside members alike — built as a new module inside the existing BrainBuilder ERP.

PROJECT	DATE
Brain Builder International – Backend API	15 Jul 2026
PLATFORM	STATUS
BrainBuilder ERP – Node.js / MongoDB	Draft for review

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01 Vision & scope

The system is not a "book issue register." It is a general-purpose **resource library** — any physical item the institution wants to lend, rent or sell can be added to the catalog: books, toys, STEM kits, lab equipment, musical instruments, sports gear, stationery. The catalog structure is category-driven, so new item types can be introduced later without touching the code.

Two audiences share the same catalog and the same rules engine, but see different storefronts:

- **School side** — students, teachers and parents, tied to the existing student/class/parent records already in BrainBuilder.
- **Public side** — anyone outside the school who registers as a library member and buys, rents or borrows against a paid membership.

Everything — pricing, fine rules, who can approve what, which items are issue-only vs. rentable vs. for sale — is **configured by the admin**, not hardcoded.

02 Who uses it

Access is role-based, matching the roles the platform already has (`admin` , `sub-admin` , `teacher` , `parent`), plus two new participants.

EXISTING

Super Admin

Owns every setting: categories, pricing, fine rules, delegation. Can see/override any transaction.

EXISTING

Librarian (Sub-admin)

Staff given a slice of admin power, e.g. "catalog + issue/return, no pricing." Same delegation model as today.

EXISTING

Teacher

Optional: reserve/issue class-set items if granted. Read-only on class usage otherwise.

EXISTING

Parent

No new login. Same account, now with a Library tab showing every child's activity.

EXTENDS

Student

Browses catalog, reserves items, sees own due dates/fines — directly or via parent app.

NEW

External / Public Member

Self-registers, no school record needed. Buys, rents or borrows against a membership plan.

03 Catalog & inventory

Every item lives under a **category** → **sub-category** tree the admin defines, so books, toys and materials sit in one catalog instead of three separate systems.

Two levels of record, on purpose

A **title** (e.g. "Encyclopedia of Science, Vol. 2") is the catalog entry. Each physical unit is a **copy**, with its own accession/barcode number, condition and current status — because you can own five copies of one book and need to know which is with which student.

Core fields captured per title / per copy

FIELD	LEVEL	PURPOSE
Title, category, sub-category	Title	Catalog placement
Author / brand / publisher	Title	Search & filtering
Suitable age / grade	Title	Relevance to the right student
Allowed modes	Title	Issue-only, rentable, for sale, or a mix
Price, rent-per-day, deposit	Title	Feeds the payments engine
Accession / barcode no.	Copy	Unique physical identifier
Shelf / rack location	Copy	Physical retrieval
Condition	Copy	New → Good → Fair → Damaged → Lost
Status	Copy	Available, Reserved, Issued, Under repair, Retired, Sold

Low-stock/reorder thresholds are configurable per title.

04 Membership models

SCHOOL MEMBER

Linked automatically to the existing student/teacher record — no separate signup. Limits can differ by class/grade. Charges can route through the same fee/wallet ledger already used for tuition.

EXTERNAL MEMBER

Self-registers with ID proof (reuses existing document-upload flow). Chooses a plan, pays a refundable deposit, gets a member ID before borrowing/renting begins.

Admin controls both: limits, fees, deposits and validity periods are settings, editable without a code change — independently for school and public members.

05 How an item moves

Every copy carries a status that only moves along allowed paths — this is what stops double-issues and lost items from silently vanishing from the count.

Available → Reserved → Issued → Returned → Available

Issued → Overdue → Fine Raised → settled → Returned

Issued → Lost / Damaged → Replacement Billed (closed)

Returned → Under Repair → back to Available once fixed

Available → Sold (ownership transferred, closed)

Three acquisition modes, set per title by the admin:

- **Issue** — free, time-boxed borrowing, typically for school members within a policy (e.g. 14 days, renewable twice).
- **Rent** — paid, duration-based, with a refundable deposit; open to school and public members.
- **Buy** — outright sale; ownership transfers, no due date, no fine.

06 Admin control & delegation

Reuses the permission engine already running the platform — a named `Permission` assigned to a `Sub-admin` record with a scope and an optional expiry date. The library module adds its own permission names to that same system.

Example permission keys the admin can hand out individually

PERMISSION	GRANTS
<code>library.catalog.manage</code>	Add / edit titles & copies, set pricing
<code>library.member.manage</code>	Approve external members, edit membership plans
<code>library.issue.manage</code>	Issue, return, renew, reserve
<code>library.fine.manage</code>	Waive, adjust or collect fines
<code>library.inventory.audit</code>	Run stock checks, mark lost/damaged/retired
<code>library.report.read</code>	View reports & dashboards, no edit rights

Exactly like the existing sub-admin model, each grant can be **scoped** (one branch, one class group) and **time-bound** — a temporary librarian for the summer term automatically loses access when it ends, with no manual cleanup.

07 What the parent sees

A read-only, always-current view inside the parent's existing portal — no separate login.

Per-child Library tab, visible to the linked parent account

COLUMN	SHOWS
Item	Title, cover image, category
Action	Issued / Rented / Bought / Returned / Renewed
Date	Exact date & time of the action
Due / return date	When it's expected back, or was returned
Amount	Rent paid, purchase price, or fine, with receipt
Status	Active, overdue, returned, lost/damaged charge pending

History is permanent — a parent can look back at everything their child has ever issued, rented or bought through the library, exactly as they already can for fee payments.

08 Self-service portal

Browse & reserve

Search by category, age group or availability; hold an unavailable item and get notified when it's back.

Pay & track

Pay rent, purchase price or a fine online; see full history and current dues anytime.

09 Notifications

Runs on the push and in-app notification system already built into BrainBuilder — the library module only triggers it on schedule.

- **Due-soon reminder** — a day or two before the due date.
- **Overdue alert** — to the member and, for school members, to the parent as well.
- **Reservation ready** — the moment a held item is back in stock.
- **Payment / fine receipt** — instantly on collection.

10 Payments & fines

- **Fine formula is a setting** — per-day rate, grace period and a maximum cap, editable by the admin, and can differ for books vs. toys vs. equipment.
- **Deposits** on rentals are held against the member's ledger and auto-refunded on a clean return; damage or loss converts the deposit into a replacement charge instead.
- **School members** can have library charges added to the same invoice/wallet as tuition fees, or kept separate — admin's choice.
- **External members** pay through an online gateway at the point of rent, purchase or fine settlement.

11 Reports & dashboard

Operational

Overdue list, active reservations, low-stock titles, items due for repair, defaulter list by class/member.

Financial

Rental income, sales revenue, fines collected, deposits held, category-wise stock value — daily, monthly or custom range.

12 Branches & audit trail

If the library spans more than one campus or branch, every title, copy and transaction is scoped to a branch, and reports can be viewed per-branch or combined.

Every action — who issued, returned, edited a price, waived a fine, or marked an item lost — is written to the same audit log already tracking the rest of the platform. Nothing is silently changed or deleted; everything is attributable to a user and a timestamp.

13 How this fits BrainBuilder today

Nothing here is a parallel system — it's new modules plugged into the platform already in place.

Reused vs. new

CAPABILITY	SOURCE
Login, roles, JWT auth	Existing User & auth module — unchanged
Delegated admin access	Existing Permission / Sub-admin models — extended
Parent ↔ student linkage	Existing Parent module — Library tab added
Push & in-app alerts	Existing notification module — new trigger events
Action logging	Existing audit module — new event types
ID / KYC documents	Existing document-upload module — reused
Catalog, copies, memberships, transactions, ledger	New — this module

14 Roadmap

CORE ships everything above. These extend it once the core is live and in use:

- PHASE 2** Barcode / QR scanning for instant issue-return at the counter
- PHASE 2** Self-checkout kiosk for students
- PHASE 2** Reading rewards / points for students who return on time
- PHASE 2** Home delivery / pickup scheduling for external members
- PHASE 2** Vendor & procurement tracking for restocking